



Strategy for Crisis/Emergency Management

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OPPORTUNITIES FOR IMPROVEMENT: SIGNS

IMPROVING SIGN MESSAGES

An inventory of current signs should be made to assess what signs exist, their condition and relevance. This audit could also identify who is posting the signs; once known, groups posting signs, as well as the messages that they are posting, can be more easily coordinated. Where there are too many messages displayed, consider minimizing, consolidating or otherwise spacing the messages to diminish the likelihood of sign pollution, mixed messages and disregard of important messages.

The objective of the sign messages is to support the intended users by providing education, location, information about acceptable behaviors, directions, warnings, ownership, and access guidance. Whenever possible keep sign messages positive. For example, instead of “Don’t pick the flowers” say “Leave the flowers for the next visitor.”

Sign messages should be simple, clear and visible from a safe distance and for a reasonable period of time. The use of clear, easily understood symbols or pictograms, in place of or in addition to words, allows for faster and more accurate interpretation. When signs are for those on the river, providing enough time to interpret the message while water is flowing at its fastest speeds will maximize comprehension and compliance. On the land trail, consider the speed at which a cyclist may pass a sign to determine the appropriate size, symbol, message and placement of signs.

Consider adding signage that indicates location for a rescue response. Using help numbers on the water trail to identify location should also serve to identify corresponding locations on the land trail. It is not recommended that independent trail mile markers be used as the help numbers on the land. Duplication of mile markers on various

trails throughout the heritage area could cause confusion with emergency response teams and possibly delay emergency response to an incident.

Land trail markers are recommended in addition to help numbers, as walkers and bikers tend to be interested in distance traveled.

Color and shape coordination of messages provides users with a quick and easy reference. Just as red signs signal a higher level of action than a yellow sign on our roadway system, so too can such color (or shape) indicators help us quickly and easily provide important messages to the users of the trail system.

Consider providing reflective type/symbols on signs that provide guidance to emergency responders. This will provide emergency responders improved wayfinding during hours of darkness. Be certain that the reflectivity of the sign does not blind the responder.

Guidelines for messages should be developed for all interested/responsible parties to follow when developing new signage. Such guidelines would increase consistency, allowing for greater ease of use, greater compliance of message and reduced liabilities. If implemented within a comprehensive sign strategy, it will also likely save lives!

Consider adding a message on signs at trailheads that encourages visitors to carry cell phones. Though the coverage may depend on the carrier and placement of antennas, national standards allow 911 calls on any carriers system without charge. Working with the local cell phone carriers can help encourage greater coverage.

Signage utilizing geographic nomenclature can be utilized to indicate available communication resources. Fire stations that use place names and nearby downtown areas all offer opportunities for communicating a crisis situation.



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With the river being so long, and passing through so many different jurisdictions, it is imperative that signage indicate a correct location. Consideration should be given to have the signs indicate which jurisdiction (county/township/borough) the callers are located in. A frequent problem with cellular phone calls to 911 centers is that the call may be routed to a dispatch center far removed from the actual incident location unbeknownst to the caller. For example a caller may connect to the 911 center and advise that an incident is occurring on the river across from the “white house with the big porch and just downriver from the boat ramp.”

IMPROVING SIGN PLACEMENT

The establishment of placement guidelines will help to minimize sign and message pollution and lead to more readable, understandable and enforceable messages.

Developing the sign placement strategy can provide an opportunity for all stakeholders to contribute. Once the strategy is jointly agreed to, uniformity and conformity can be achieved. All three user groups (intended trail users, unintended trail users and emergency response personnel) must be considered in this process.

Sign locations for emergency response personnel should serve to clearly guide responders to affected areas and or persons. When someone dials 911 for help and identifies his or her location along the trail (like the area posted in photo 5) the 911 operator should be able to determine the best vehicular access point, land trail access and river access points for emergency responders. These signs should also warn of restrictions; for example, if a fire truck could not safely travel a land trail due to a lack of turnaround space or sufficient trail width.

Observing the unintended uses of the trail also helps in determining sign locations. By assessing where graffiti is prevalent, we can ascertain

two things: the first is that these are the areas where signs are highly visible; the second is that the unintended users have claimed these areas of the trail and that signs will have to be positioned and constructed to both discourage the current unintended uses of the area and protect the signs.

Sign placement for crisis/emergency management considerations should include:

- **Trail heads** – access guidance, public appeal, signs of ownership, appropriate warning signs, expected and acceptable behaviors, and location
- **Any parking or designated rest area** - access guidance, public appeal, signs of ownership, appropriate warning signs, expected and acceptable behaviors, and location
- **At road intersections preceding trail heads** – directional, access guidance
- **At mile intervals along the land trail** – mile/location markers, public appeal
- **At water trail tributaries** – location markers
- **At loop and wayside trails** – location markers, distance signage
- **Preceding dangerous areas** – location markers, warning signs, public appeal
- **At dangerous areas** – location markers, warning signs, public appeal

As noted earlier, it is crucial that warning signs be located so the viewer (usually in motion) has time to avoid the danger. Placing warning signs several hundred yards before dams or other hazards would greatly benefit the river users; such signage should be designed to function when



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water is fast moving. Signage properly placed upriver from dams could contribute to lowering the risks of injury and death. Signage at landings could include indicators of the nearest river hazards. On land the trail signs that precede hills and sloped intersections frequented by bicyclists should be placed early and often enough to provide sufficient warning of impending dangers.

Signs warning of an approaching dam (or any other potential danger) should be in place in all locations along the river where boaters may encounter such dangers. Using the Black Rock Dam as an example – the Route 113 Bridge is located approximately 1.25 river miles upstream of the dam and would be an ideal location for warning signs advising of the approaching dam. The signage could be painted on the bridge superstructure in reflective type paint or a sign panel could be attached to the bridge. Placing the warning sign on the bridge would reduce the chance of removal or defacing by vandals.

ENSURING PROPER SIGN MAINTENANCE

Installing signs along a trail is only the beginning of an ongoing process. Once installed, the signs will need to be surveyed and maintained regularly to insure appropriateness of message, location and condition.

Over time the location in which the sign is installed may take on different meaning and uses. New user groups may become active while others become inactive. Maintaining the appropriateness of the sign messages will require an annual (at least) investment in time and resources.

Monitoring the condition of signs is also important. One method is to encourage trail users to report problems. Other policies must be put in place to spot problems and prevent a backlog of repair and maintenance of signage.

Consider a program to survey trail signs on a yearly basis. Such a sign audit program should include appropriateness of message displayed, visibility of the sign (looking for the encroachment of vegetation) and the condition of the sign (looking for decay and vandalism). The auditor should be supplied with a sign location map so that missing signs can be identified.

Enlisting the help of local youth clubs in the audit can further build interest and territoriality as well as teach lessons pertaining to graffiti and vandalism. Conducting such audits prior to the heaviest times of use is the most beneficial.

PREVENTING INSUFFICIENT SIGNS

One of the reasons that sign guidelines are needed is to prevent insufficient signs in the future. Once guidelines have been established for what sign types are needed, what messages need to be communicated, where signs should be placed, how they should be designed, and how they will be constructed and installed, then it can be clearly determined where signage is required.



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OPPORTUNITIES FOR IMPROVEMENT: TRAILS

ADDRESSING SURFACE INCONSISTENCY

Indicate where surface changes are about to occur. Provide enough advanced warning for cyclists.

ADDING FEATURE LOCATORS

Trail features should be marked with signage indicating use, location and special features. Consider landing signage and surface reinforcement.

UTILIZING TRAIL VEGETATION

Consider utilizing low growing, dense foliage on each side of the trail to discourage hiding and other illicit activity. Ideally, an eight-foot "shoulder" (on each side of the trail) would increase the total viewable width to 24 feet. Otherwise consider mowing or pruning vegetation to the edge of the right-of-way.

ADDRESSING TRASH REMOVAL

A permanent and consistent trash monitoring and removal program would benefit all trail users and the communities that border the trail system. Removal of dangerous material makes the trail safer. A clean trail system makes vandalism and criminal activity more apparent and encourages awareness of suspicious activity. Work with volunteer groups to encourage reporting and or removal of dangerous river debris. Only trained personnel should be responsible for removal of debris near hydraulics and other noted dangerous areas along the trail.



INSTITUTING TRAIL REPAIR

There are many civic organizations that could be approached to make trail repair part of their annual project list. Scout troops and other civic organizations often look for such opportunities. Fence repair, erosion prevention and similar projects would benefit the trail system and foster a greater partnership with the trail association. It is understood that Fairmont Park is responsible for the Flat Rock Dam Area.

IMPROVING GRAFFITI REMOVAL

Graffiti that exists in the area of the trail will serve to discourage legitimate use of the trail system. Removal of such unintended signage will help to maintain the aesthetics and appropriate use of the trail system. Removal of graffiti within five working days has become a national standard (this policy results in a maximum of seven days). Working with the agencies that maintain structures on which graffiti is placed can help create policies and response teams to address the issue.

Photos of tagging and suspected gang graffiti are generally recommended prior to the removal or covering. Photos can be a great investigative tool since taggers use signature markings.

IDENTIFYING HAZARDS

Hazards need to be identified in light of the guidelines for sign placement and messages noted above. Consider modifications to the trailheads that cross roadways to insure a ninety-degree angle to maximize opportunities for natural and convenient surveillance. Bicyclers will find it easier to look right and left at that angle.



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OPPORTUNITIES FOR IMPROVEMENT: STEWARDSHIP

FOSTERING A SENSE OF OWNERSHIP

It is important to reach out to stakeholders to become involved in the life of the trail. Youth programs might care for the signs in the areas they frequent; agencies or groups who are posting messages could also be enlisted in sign maintenance.

Due to hesitation on the part of many of those interviewed to involve local companies in such programs as “Adopt-a-Trail,” perhaps a program that invites the involvement of groups already using the trails would be a better approach. A by-invitation-only Trail Watch program is a viable method to encourage territoriality by established groups that already use the trail system. The goal is to further reinforce each person’s relationship to the river and land trails and encourage his or her stewardship.

Although there is a desire to create a fully conforming sign strategy for the Schuylkill River Trail System it could diminish the sense of ownership, or territoriality, of local stakeholders. Territoriality is created by those who feel a sense of ownership and, due to that ownership, provide protection for an area. If only one group has control it is very difficult to provide adequate territoriality/protection for the entire length of the land and water trail. A good compromise is to involve key stakeholders in the development of the sign strategy.

PARTICIPATING IN TRAIL MAINTENANCE

The need for stewardship is paramount.

Although not an “Adopt-A-Trail” program, targeting club sponsorship for trail segments they already use could result in territorial reinforcement, maintenance, monitoring of conditions and reporting of suspicious activity. Due to the close proximity of local colleges, perhaps the students and staff

could be brought together to watch over neighboring areas. It is possible that the river could also tie into some of their studies, such as biology or ecology

ADDING INTERPRETIVE AMENITIES

Geographic nomenclature can tap into local history and pride and should be taken advantage of to involve groups and communities. An event or person immortalized in a street name could be further recognized with a trailhead marker or interpretive sign. This would help garner community interest.

Consider taking an inventory of historical and cultural resources. This may result in many more community tie-ins that can be taken advantage of in geographically significant trailhead names and river landings. Marking trailheads with local geographic nomenclature may result in greater local interest; at the very least it will make locating trailheads and river landings that much more logical and easy.

The Schuylkill Area Heritage Area should be visible and related to significant places, events and names.

SOLICITING COMMUNITY SPONSORSHIP OF TRAIL FEATURES, TRAIL HEADS AND TRAIL ROUTES.

Such commercial buy-in can result in extended financial resources as corporations may seek tax deductions from non-profit foundation interaction.



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DEFINING THE TRAIL PERIMETER

Each land trail planned must serve a specific purpose, community, group and/or neighborhood. It should not be assumed that building a trail will lead to appropriate and effective use.

If the perimeter of a trail is close to a neighborhood or community it should be designed to be very visible; this approach connects it to the community and allows for natural surveillance.

A lack of a visual definition between spaces can result in an increase in community crime not directly related to the trail system. Unintended trail users look just like intended trail users even as they move off the trail to burglarize homes and businesses bordering the trail.

Consider a change in elevation or standardization of surfacing materials as a way of creating edge definition. Movement up an elevation from a fully public sidewalk to a semi-private front yard, for example, makes it clear you are moving into someone else's space. Crossing over a change in texture (driving from a smooth public road across cobblestones into a private parking lot) lets you feel the transition. The same can be done with curbs (be sure to include ramps for persons with disabilities), painted lines, lighting gradient changes and signage. In the case of a trail an elevated trail makes it obvious that someone is doing something wrong when he or she is at a lower elevation than the trail. That person deserves increased scrutiny.

Perimeter vegetation can create a reinforced perimeter and separate trail easements from other uses. Low, self-limiting or "self pruning" vegetation that is applied consistently throughout the land trail system could become part of the branding of the trail system and help to identify it. Consider that someone who does not look like a trail user (clothes, movement direction, mannerisms) is standing near the perimeter (as

identified by signage and vegetation) and is therefore more conspicuous by their mere presence.

SUPPORTING NATURAL SURVEILLANCE:

We need to create environments where there is plenty of opportunity for people engaged in their normal behavior to observe the space around them. Most criminals do not want to be noticed while committing crimes. Areas can be designed so they can be more easily observed.

For example, residential homes that abut sections of the trail can be provided clear lines of sight to the trail. These same neighbors can be enlisted to watch over the trail segment and report and suspicious behaviors to the proper authorities. Providing these neighbors with appropriate numbers of who to call, what to look out for and explaining that their input is important will all serve to make the trail a safer place. The same tactics can be used with local businesses and user groups. Such community activism tends to increase usership as well.

Another option is to maintain a fairly isolated trail so that trail users do not get the feeling that they are in a populated area. This runs counter to most crime prevention programs but may well fit the needs of the trail users in some cases. If a degree of isolation is desired there are other measures that can be taken to insure the safety of trail users.

PROVIDING INDICATORS OF PLACE

Providing interpretive signage along with geographic names can provide identifiable trailheads, trail perimeters and boost community interest. Trail users are, by common nature, storytellers.

Providing interpretive signage along with geographic names can provide identifiable trailheads, trail perimeters and community interest. People stopping to read an interpretive sign can be directed to look in a particular



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direction to view sites beyond the sign or described feature. This “coerced” observation is part of the aforementioned concept of natural surveillance.

COMMUNICATING TO THE PUBLIC

Advertising to users how the sign strategy works will help spread awareness of trail benefits. It will serve to help users feel safer, more in tune with their surroundings and more prepared in the event of an incident/crisis. Advertising can be accomplished on the web sites, through brochures provided to local stores (such as local sporting goods stores), libraries, visitor’s bureaus, etc. The brochure can show the different colors and symbols along the trail. A walker can then become familiar with the concept that there are mile markers along the land trail and how to explore new side trails and still count the distance covered. A group of school children could be provided a map showing the locations of the educational markers along the trail. Cyclists could look for areas where the trail crosses other roads, so that they can be better prepared to stop, and look both ways, prior to finding themselves in a dangerous situation. Boaters can be educated on appropriate landings, location signs, camp sites, warning signs and how to read water height and speed signs to interpret their level of safety. Simple brochures with maps and keys to these types of signs have numerous benefits.

Posters that show the different types of signage could also be posted in various locations throughout the neighboring communities and at trail heads and river landings.

Web sites should be reviewed periodically to maximize user-friendliness and minimize conflicting or outdated information.



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OPPORTUNITIES FOR IMPROVEMENT: EMERGENCY RESPONSE CAPABILITIES

EMERGENCY RESPONSE CAPABILITIES

Add location signage at regular intervals. To minimize the potential for delayed responses it is vital to have specific signage at regular locations indicating where the incident is occurring. Additional time spent looking for the incident increases the time it takes for rescue personnel to arrive on the incident scene and reduces the possibility of a successful rescue.

Reserve boat launch and trail access points for emergency personnel. Similar to “no parking except for emergency vehicles” signs seen at shopping centers and schools, these signs would note locations where emergency responders could access the trail or launch a boat into the river. Fire departments currently do have predetermined launching sites for boats.

Make some sites off-limits with barriers and signs. A local responder suggested some sort of physical barrier on the dams would help to keep people from falling off or gaining access to the dams.

Involve response agencies in the planning process. Involving all the different jurisdictions and response agencies in the master sign plan process will not only increase the success of the program, but also reduce issues between the agencies and jurisdictions when incidents do occur. For example, if developing ground signage that must be read by helicopters, it would be helpful to solicit input from individuals experienced in helicopter operations.

Formalize aid agreements among all response agencies. Agencies and departments with emergency response responsibilities along the river and

trail should have written automatic/mutual aid agreements to reduce liability issues when an incident does occur.

Continue joint training. Regular training sessions with the participation of neighboring jurisdictions is crucial.

Develop written policies for water rescue operations. These procedures should be made available in the form of a preplan document to all departments, and should be used during training exercises and simulations so that they are familiar to response personnel. In addition, they should be reviewed and revised annually.

Coordinate radio frequencies. Consider providing radios that share frequencies to agencies along the river and trail. For example, personnel at the Schuylkill Canal Association’s recreation area would then have the ability to communicate quickly and directly with emergency responders, especially due to the number of incidents that occur in this area. Providing radios to the local police departments, and Pennsylvania Fish and Game Commission personnel in the area should also be considered.

Be aware of available resources. For example, helicopters are available from a number of different agencies, such as Penn Star and the United States Coast Guard. The helicopters primarily do recon work for the land-based units, coordinating and directing these units to the locations of victims. Landmarks and signs are helpful for the pilots to direct land-based personnel.

Conduct a public education program. As with other programs designed to prevent or reduce deaths and injuries, an educational component is critical to the success of the overall plan. The cable television program on water rescue mentioned earlier is an example of how to educate the public about the signs and how they can help in an emergency situation.



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SUMMARY OF SIGN TYPES NEEDED:

SIGNS FOR CRISIS/EMERGENCY MANAGEMENT:

Location information signs, which allow users to identify their current location to others in the event of an emergency.

Expected and acceptable behavior signs, which communicate acceptable activities, behaviors and actions and those that are not acceptable. Such signs provide a deterrent to potential rule breakers and allow those not complying to stand out, be reported and dealt with.

Directional signs provide directions on how to get from one location to another. On land, such signs can provide directions to towns, local sites, areas of interest, trail spurs, etc. On the river, directional signs can indicate safe passage around dams, bridges, and locks. Landings as well as camp sites can also be identified.

Warning signs can identify changes in surfacing materials or elevations on the land trail, while on the river can include water level, water speed, and dangerous areas (dams, locks, bridges, rapids, etc.).

Access guidance signs clearly identify the land based trail heads, branches of the trail system, river trail landings (a.k.a. take-outs), campsites, and exits to neighboring areas of interest, etc. These signs encourage increased usage, minimize confusion over trail location, provide safer passage, and build trail awareness.

Temporary signs can be used to replace a damaged or missing sign, provide additional information during an event, divert traffic around a section of trail where there may be a fallen tree, mud slide, accident, or fire, in addition to any number of other purposes.

SIGNS THAT FOSTER STEWARDSHIP:

Public appeal signs direct legitimate users and neighbors of the trail on how and where to report problems (as well as what types of problems to report) and usually include a phone number.

Educational signs describe the historical significance of trail features and encourage greater use of the trail by local school groups, history buffs and other interested parties. These signs can be supplemented by print brochures and web-based information.

Signs of ownership create a sense of pride and responsibility. When a trail has neighborhood identification signs, individuals and groups are more likely to care for areas and signs to which they have a connection. This has the additional effect of encouraging other users to take similar care of the trails.



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APPENDIX

CRISIS/EMERGENCY MANAGEMENT TERMINOLOGY

Mitigation, the strategy of reducing the likelihood that an incident will occur. Mitigation sets the stage for crisis/emergency planning by providing trail users the guidance needed for safe use of the area, guiding users to appropriate areas and helping them avoid dangerous ones, encouraging stewardship and upkeep, identifying dedicated pathways, communicating areas that pose a risk, and directing trail users on how to report problems.

Preparedness, the ability to react to emergency situations. Its effectiveness can be maximized by practicing for response and recovery situations and by preparing emergency response agencies.

Response, the crisis/emergency management strategy put into action during actual events.

Recovery, the return of an area to 100% operational status following an incident.

Temporary signage, used during recovery to guide users away from affected areas of the trail that may pose a risk. It is important that this signage be well-maintained, as worn signage conveys diminished importance of the message.

Continual evaluation, the ongoing assessment of sign condition (placement and message) and emergency responses observed during practice drills and events. This provides a clear picture of the system's effectiveness and the ability of those responsible for handling emergencies.